Modeling with Partial Differential Equations
Exercises - Session 3: the Heat Equation

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Exercise 1 — Analytical solution of the one-dimensional heat equation We aim to solve the Cauchy problem

$$\begin{cases} \frac{\partial u}{\partial t}(t,x) = k\frac{\partial^2 u}{\partial^2 x}(t,x), & t>0, \quad x\in(0,L), \\ u(0,x) = u_0(x), & x\in(0,L), \\ u(t,0) = u(t,L) = 0, & t>0, \end{cases} \tag{1}$$

where $k>0$ is the thermal diffusion coefficient and $L>0$ the length of the medium in which heat diffuses. To this end, we use the method of separation of variables.

1. **Separation of variables**
   Seek a solution of the form $u(t,x) = T(t)X(x)$, for an initial condition u_0 which is not zero. What does the partial differential equation become when u is replaced by this product? Show that there exists $\lambda\in\mathbb{R}$ such that
   $$X''(x) + \lambda X(x) = 0$$
   and
   $$T'(t) + k\lambda T(t) = 0.$$

2. **Equation in x (Sturm–Liouville)**
   The spatial solution $X(\cdot)$ must satisfy, for some λ,
   $$X''(x) + \lambda X(x) = 0, \qquad X(0) = 0, \quad X(L) = 0. \tag{2}$$
   For which values of λ (eigenvalues) does (2) admit nontrivial solutions $X(\cdot)$? What are those solutions (eigenfunctions)?

3. **Equation in t**
   For an eigenvalue λ_n, give the differential equation satisfied by $T_n(t)$. What is the associated family of temporal solutions $T_n(\cdot)$? Deduce a family of solutions to (1) for well chosen initial conditions u_0.

4. **Finite sum solution**
   Suppose $u_0(x)$ can be approximated or given exactly by a finite sum of sines
   $$u_0(x) = \sum_{n=1}^{N} u_{0,n} \sin\left(\frac{n\pi x}{L}\right),$$
   with a fixed integer N, where $(u_{0,1},\dots,u_{0,N})\in\mathbb{R}^N$. Write the general solution $u(t,x)$ of the problem as a sum of sine functions that satisfy the initial condition.

   Remark: Although we only work with finite sums here, one can also make sense of the solution when the sum becomes infinite, leading to what is called a Fourier series.

Exercise 2 — Uniqueness of solutions to the heat equation Let $T>0$, $k>0$, and $f\in C^0([0,T]\times[0,L])$ be fixed. We aim to show that for any initial datum $u_0\in C^0([0,L])$, there exists at most one solution $u\in C^{1,2}([0,T]\times[0,L])$ to the problem

$$\begin{cases} \partial_t u(t,x) = k\partial_{x,x}u(t,x) + f(t,x), & (t,x)\in(0,T)\times(0,L), \\ u(0,x) = u_0(x), & x\in(0,L), \\ u(t,0) = u(t,L) = 0, & t\in(0,T). \end{cases} \tag{3}$$

1. Using the linearity of the heat equation, show that uniqueness for (3) is equivalent to the statement that the zero solution $w \equiv 0$ is the only solution to the homogeneous problem:

$$\begin{cases} \partial_t w(t,x) = k\partial_{x,x} w(t,x), & (t,x) \in (0,T) \times (0,L), \\ w(0,x) = 0, & x \in (0,L), \\ w(t,0) = w(t,L) = 0, & t \in (0,T). \end{cases} \tag{4}$$

2. Set $E(t) = \frac{1}{2}\int_0^L w^2(t,x)\,dx$ for all $t \in [0,T]$, where $w \in C^{1,2}([0,T] \times [0,L])$ is a solution of (4). Show that $E \in C^1([0,T])$ and that

$$\forall t \in (0,T), \quad E'(t) = \int_0^L w(t,x)\,\partial_t w(t,x)\,dx. \tag{5}$$

3. Show the following identity

$$\forall t \in [0,T], \quad E'(t) = -k\int_0^L \left(\partial_x w(t,x)\right)^2 dx, \tag{6}$$

and then conclude.

4. Return to equation (3), and assume $g = 0$.
Let us admit the Poincaré inequality: for all $h \in C^1([0,L])$ such that $h(0) = h(L) = 0$, we have

$$\int_0^L h(x)^2\,dx \leq \frac{L^2}{\pi^2}\int_0^L \left(h'(x)\right)^2 dx. \tag{7}$$

Using the same reasoning as in the previous question, show that

$$\forall t \in [0,T], \quad \int_0^L u(t,x)^2\,dx \leq e^{-\frac{2k\pi^2}{L^2}t}\int_0^L u_0^2(x)\,dx \tag{8}$$

and deduce the convergence rate of the solution.

Exercise 3 - A proof of the one dimensional Poincaré's inequality We recall that $C_c^1((0,1))$ stands for the set of all continuously differentiable functions with compact support in the open interval $(0,1)$.

Theorem 0.1 (One-Dimensional Poincaré Inequality)**.** *Let $\phi \in C_c^1((0,1))$. Then the following inequality holds:*

$$\int_0^1 |\phi'(x)|^2 dx \geq \pi^2 \int_0^1 |\phi(x)|^2 dx. \tag{9}$$

This inequality is fundamental in the calculus of variations, analysis, and partial differential equations. In this exercise, you will prove it by following an elegant argument that connects it to a simple variational problem. This proof has been given by Georgios Psaradakis in [1].

Consider a given test function $\phi \in C_c^1((0,1))$ and define an auxiliary function v such that

$$\phi(x) = v(x)\sin(\pi x). \tag{10}$$

Our goal is to show $I := \int_0^1 \left((\phi'(x))^2 - \pi^2\phi^2(x)\right) dx \geq 0$.

1. Recall briefly why v is well-defined on the entire interval $(0,1)$.

2. Show that the full integrand can be written as:

$$(\phi')^2 - \pi^2\phi^2 = (v')^2\sin^2(\pi x) + \frac{1}{2}\pi\left(v^2\right)'\sin(2\pi x) + \pi^2 v^2\cos(2\pi x) \tag{11}$$

and deduce an expression of I.
Hint: Use trigonometric identities.

3. Show that, in fact,

$$I = \int_0^1 (v')^2\sin^2(\pi x)dx, \tag{12}$$

and deduce the desired Poincaré inequality (9).

Exercise 4: Thermal phase shift in a wall – Heat equation in periodic regime *Context and position of the problem.* We seek to model the propagation of daily temperature variations inside a wall. The wall is treated as a homogeneous medium occupying the half-space $x \geq 0$, where x denotes the depth in the wall. The temperature $T(t, x)$ satisfies the heat equation

$$\frac{\partial T}{\partial t}(t, x) = \alpha \frac{\partial^2 T}{\partial x^2}(t, x), \qquad t > 0, \quad x > 0, \tag{13}$$

where $\alpha > 0$ is the thermal diffusivity of the material, expressed in $\mathrm{m}^2\,\mathrm{s}^{-1}$.

The temperature at the outer surface varies periodically according to

$$T(t, 0) = T_m + A\cos(\omega t), \qquad \omega = \frac{2\pi}{\tau}, \tag{14}$$

where T_m is the mean temperature, $A > 0$ the amplitude of the oscillations and $\tau = 24\,\mathrm{h}$ their period. We are interested in the established periodic regime: the solution is periodic in time and remains bounded as $x \to +\infty$.

Part 1: A periodic solution to the heat equation

We set $u(t, x) = T(t, x) - T_m$ and we seek the harmonic regime in the form

$$u(t, x) = a(x)\cos(\omega t) + b(x)\sin(\omega t), \tag{15}$$

where a and b are two real functions.

1. Show that
$$\alpha a'' = \omega b, \qquad \alpha b'' = -\omega a.$$
What conditions do a and b satisfy at $x = 0$? *Hint: Substitute* (15) *into the heat equation* (13) *and identify the coefficients of* $\cos(\omega t)$ *and* $\sin(\omega t)$.
We set
$$k = \sqrt{\frac{\omega}{2\alpha}}.$$

2. By eliminating b from the previous system, show that
$$a^{(4)} + 4k^4 a = 0. \tag{16}$$

3. Verify that any bounded solution equation (16) can be written
$$a(x) = \mathrm{e}^{-kx}\big(C\cos(kx) + D\sin(kx)\big).$$
Hint: Show that the solution to (16) *is a vector space of dimension* 4. *Look for solutions of the form* $a(x) = \exp(rx)$.

4. Show that
$$T(t, x) = T_m + A\mathrm{e}^{-x/\delta}\cos\left(\omega t - \frac{x}{\delta}\right)$$
where
$$\delta = \sqrt{\frac{2\alpha}{\omega}}$$
is called the thermal penetration depth. Deduce that
$$\delta = \sqrt{\frac{\alpha\tau}{\pi}}.$$
Hint: Use the relation between a *and* b, *then the conditions imposed at* $x = 0$.

Part 2: Thermal attenuation and phase shift

We consider a depth $L > 0$.

5. Show that the amplitude of the temperature oscillations at this depth equals
$$A_L = A\mathrm{e}^{-L/\delta},$$
and deduce the attenuation factor A_L/A.

6. How does A_L/A vary when the thickness L increases? When the thermal diffusivity α decreases? Interpret these results physically.

   At the surface of the wall, a maximum of temperature is reached when

$$\omega t = 2k\pi, \qquad k \in \mathbb{Z}.$$

7. Show that at depth L, the same maximum is reached with a delay

$$\Delta t = \frac{L}{\delta\omega}.$$

   Using the expression for δ, establish the formulas

$$\Delta t = L\sqrt{\frac{1}{2\alpha\omega}} = L\sqrt{\frac{\tau}{4\pi\alpha}}.$$

   How does the phase shift vary when α decreases?

Part 3: Comparison of materials and sizing

We consider a wall of thickness $L = 20\,\text{cm}$ subjected to external variations of period $\tau = 24\,\text{h}$. Two materials are considered:

$$\begin{aligned} \text{dense calcium silicate block (Silka type):} &\quad \alpha_1 = 5{,}0 \times 10^{-7}\,\text{m}^2\,\text{s}^{-1}, \\ \text{rigid wood fibre board (STEICOspecial dry type):} &\quad \alpha_2 = 1{,}5 \times 10^{-7}\,\text{m}^2\,\text{s}^{-1}. \end{aligned}$$

These diffusivities are representative rounded values.

8. For each of the two materials, calculate the thermal penetration depth δ, the attenuation factor A_L/A and the time phase shift Δt expressed in hours.
   Which material best meets the objective of 12 h? Compare the attenuations obtained.

Exercise 5 — Numerical solution of the one-dimensional heat equation We aim to solve numerically the Cauchy problem

$$\begin{cases} \frac{\partial u}{\partial t}(t,x) = \alpha \frac{\partial^2 u}{\partial^2 x}(t,x), & t > 0, \quad x \in (0,L), \\ u(0,x) = u_0(x), & x \in (0,L), \\ u(t,0) = u(t,L) = 0, & t > 0, \end{cases} \tag{17}$$

where $\alpha > 0$ and $L > 0$. Introduce the grid

$$\Delta x = \frac{L}{I}, \quad x_i = i\Delta x, \quad i = 0, \ldots, I, \tag{18}$$

$$\Delta t > 0, \quad t_n = n\Delta t, \quad n \in \mathbb{N}. \tag{19}$$

We denote $u_i^n \approx u(t_n, x_i)$ and we introduce the diffusion number:

$$\mu = \frac{\alpha \Delta t}{(\Delta x)^2}. \tag{20}$$

1. Approximate the second derivative $\dfrac{\partial^2 u}{\partial x^2}(t_n, x_i)$ by a second-order centered finite-difference approximation.

2. Write the explicit forward Euler scheme associated with the heat equation.

3. Write the scheme in the form

$$U^{n+1} = BU^n, \tag{21}$$

   specifying the structure of the matrix B.

4. Show that stability in the L^2 norm of the explicit scheme requires a condition on μ, and explicit this condition.

5. Write pseudocode or a Python script implementing the explicit scheme.

6. Test numerically the explicit scheme with and without satisfying the CFL condition.

Implicit scheme

7. Write the implicit forward Euler scheme associated with the heat equation.

8. Write the scheme in the form
$$CU^{n+1} = U^n, \tag{22}$$
specifying the structure of the matrix C.

9. What is the main computational cost of advancing the solution by one time step with this method?

10. Performing a similar Von Neumann stability analysis to the one made for the explicit scheme, show that the implicit scheme is unconditionally stable.

11. Write pseudocode or a Python script implementing the implicit scheme.

12. Test numerically the implicit scheme to verify its unconditional stability.

Crank-Nicolson scheme

13. The Crank-Nicolson scheme is given by:
$$\frac{U_i^{n+1} - U_i^n}{\Delta t} = \frac{\alpha}{2}\left(\frac{U_{i-1}^{n+1} - 2U_i^{n+1} + U_{i+1}^{n+1}}{(\Delta x)^2} + \frac{U_{i-1}^n - 2U_i^n + U_{i+1}^n}{(\Delta x)^2}\right). \tag{23}$$

14. Conducting a Von Neumann stability analysis, show that the Crank-Nicolson scheme is unconditionally stable.

15. What is the order of the truncation error in time and space.

16. Implement the Crank-Nicolson scheme and compare its accuracy with both the explicit and implicit schemes.

References

[1] Georgios Psaradakis. *An elementary proof of the one dimensional Poincaré inequality and the Jacobi lemma.* Department of Mathematics, University of Western Macedonia, 2020.
Available at: `https://hal.science/hal-02873488v1`